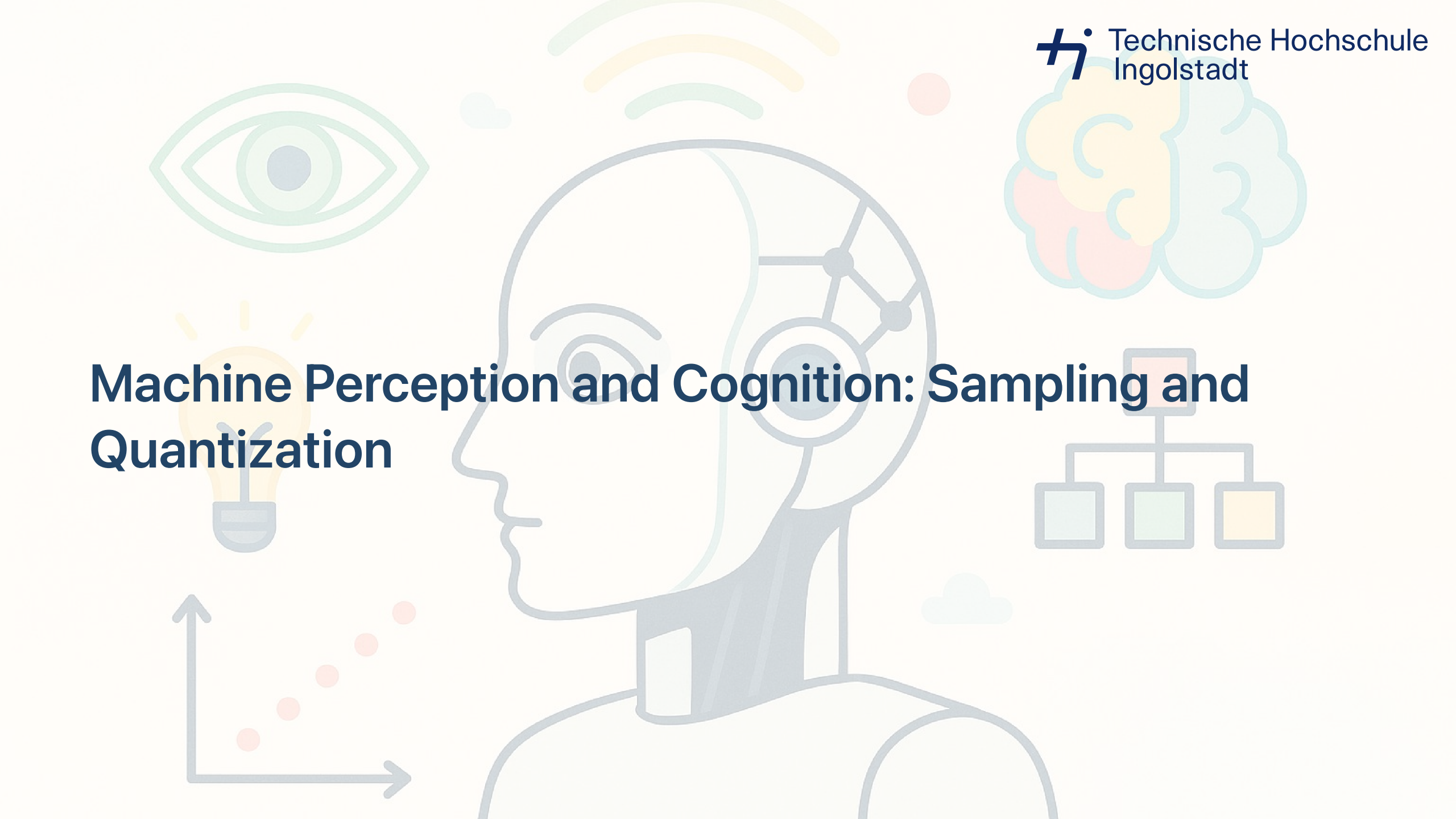
Technische Hochschule Ingolstadt



Machine Perception and Cognition: Sampling and Quantization

What is the problem?

Physical signals have the following properties:

- Continuous over time.
- Continuous over amplitude.
- Think of "infinite detail".

What is the problem?

You can zoom in on any point and still find more detail!

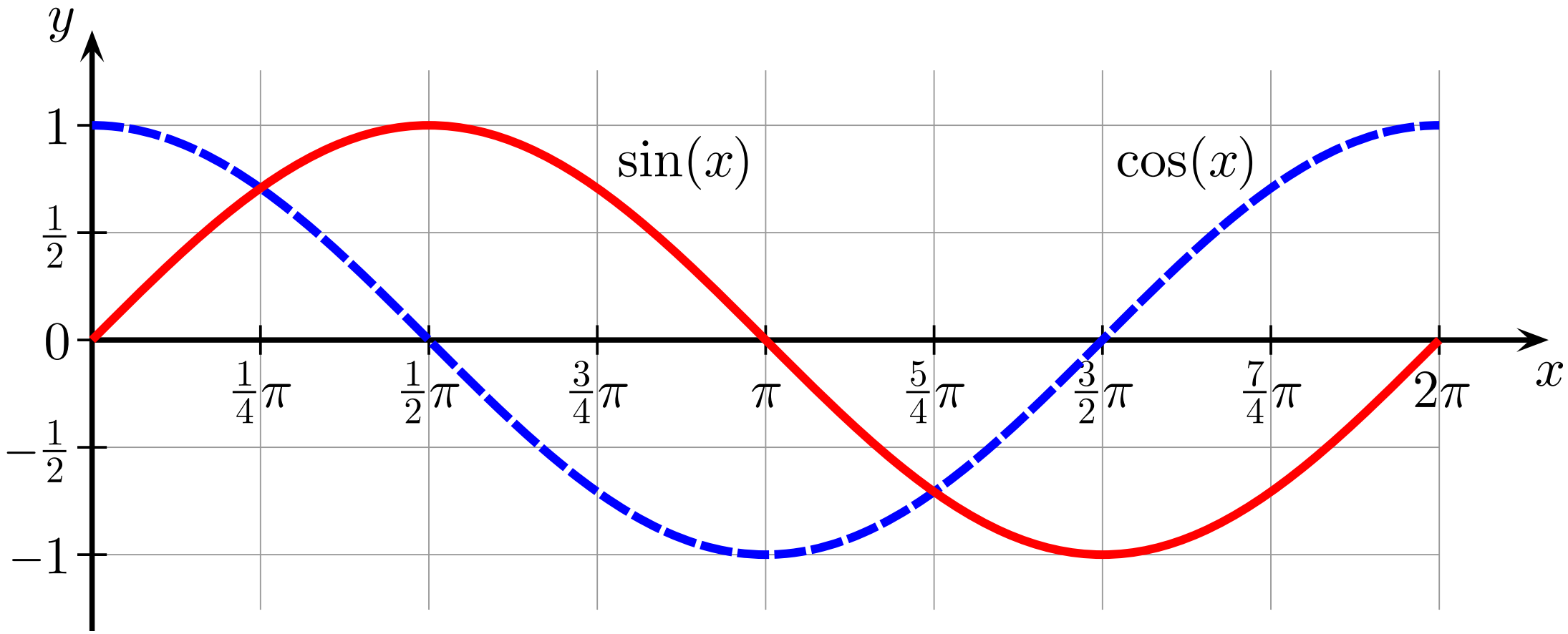


Image source: Geek3, Public domain, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Sine_cosine_one_period.svg

Fun fact!

The graphic is in SVG (scalable vector graphics), which allows to mathematically describe shapes instead of storing a rasterized image. It can therefore actually zoom and reveal more detail!

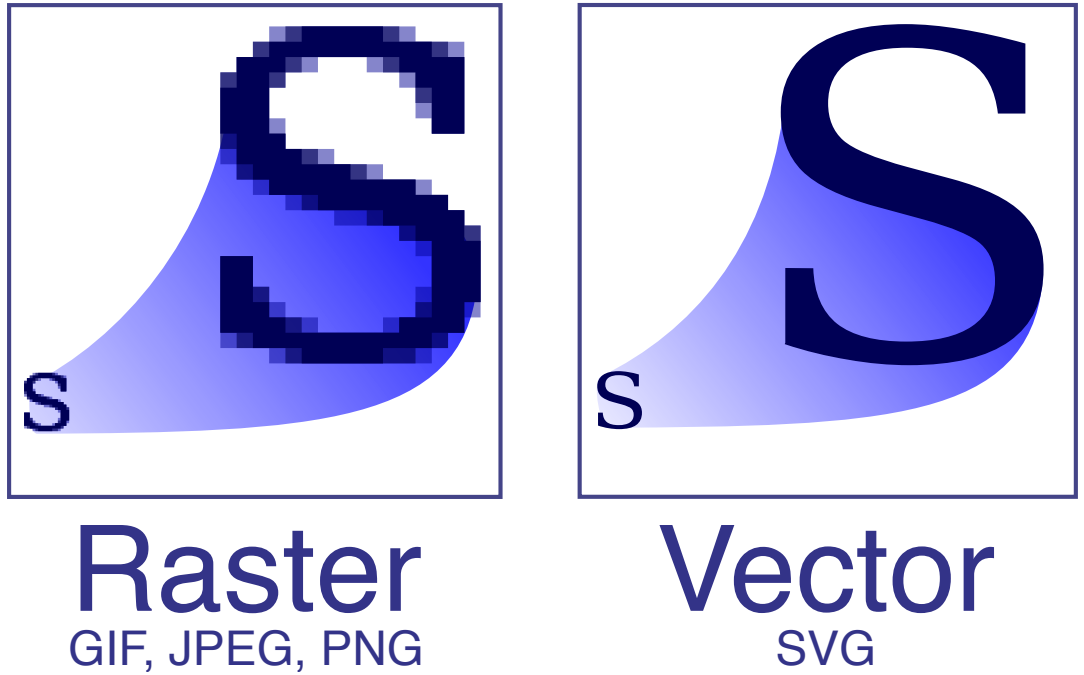
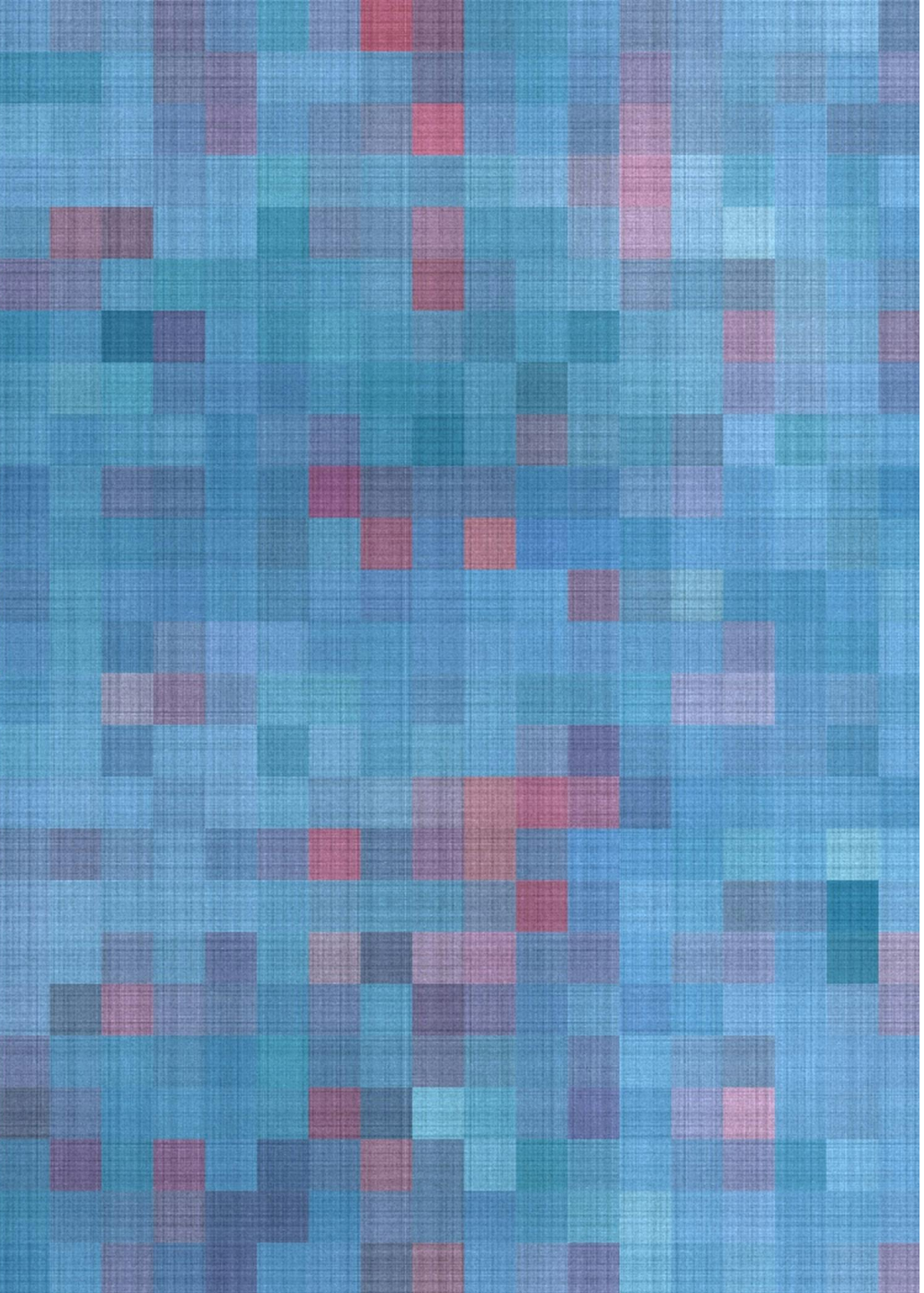


Image source: Yug, modifications by Cfaerber et al., CC BY-SA 2.5 (https://creativecommons.org/licenses/by/2.5), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Bitmap_VS_SVG.svg

What is the problem?

On the other hand, computers have the following properties:

- Limited memory (volatile and persistent).
- Limited processing power.
- Limited representation of values.



What is the problem?

Sensors also have limited resolution. Think of the number of pixels per row and column in a camera image.

Image source: https://www.pexels.com/photo/close-up-photo-of-textile-7026427/

Sampling and quantization

Sampling: Reduction of a time-/space-continuous signal to a time-/space-discrete samples.
Quantization: Reduction of an amplitude-continuous signal to discrete values.

... or ...

Sampling: Discretization in time and/or space.
Quantization: Discretization in amplitude.

Sampling

From Wikipedia, page "Sampling (Signal processing)":

> In signal processing, **sampling** is the reduction of a continuous-time signal to a discrete-time signal. A common example is the conversion of a sound wave to a sequence of "samples". A **sample** is a value of the signal at a point in time and/or space; this definition differs from the term's usage in statistics, which refers to a set of such values.

Sampling

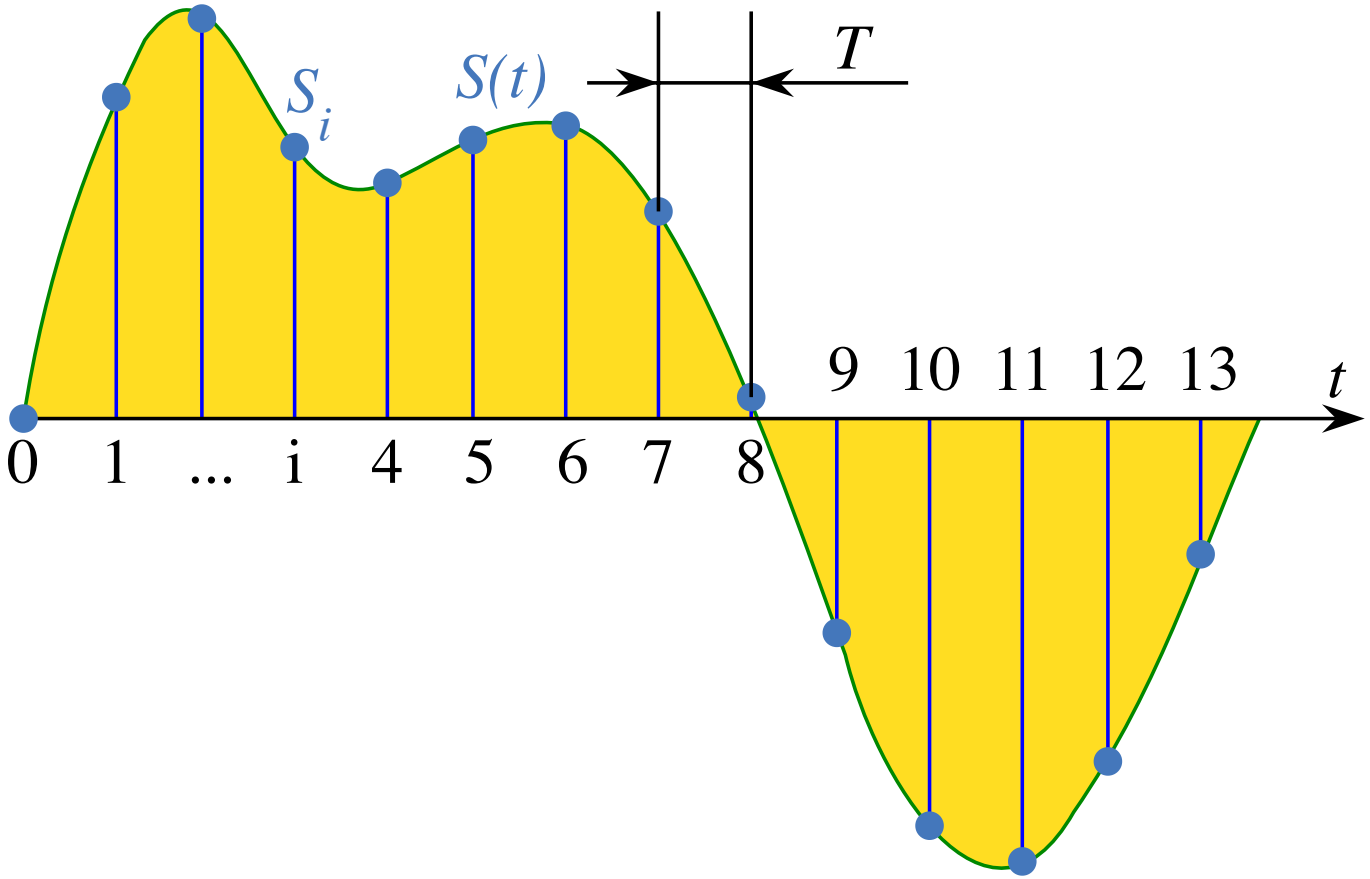


Image source: Д.Ильин, CC0, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Signal_Sampling.svg

Sampling

- Continuous signal s at time t written as $s(t)$
- In this course we restrict ourselves to uniform sampling:
  - Thus, sampling is done by measuring the signal $s(t)$ at evenly spaced intervals.
- Time interval is given in T seconds with $t_{i+1} - t_i = T$

Sampling

- Sampling frequency f_s is defined as $f_s = \frac{1}{T}$
- Sampling frequency is in units of *Hertz* (Hz), that is samples per second.

Sampling

An example:

- Sampling frequency $f_s = 100$ in Hertz.
- Sampling interval $T = \frac{1}{f_s} = \frac{1}{100} = 0.01$ in seconds.

Sampling in space

Sampling does not have to be over time only. It can also be over space, multi-dimensional space or a combination of space and time.

Think of e.g. a rasterized image!

We will later discuss potential problems with sampling and this applies to e.g. rasterized images as well.

Sampling in space

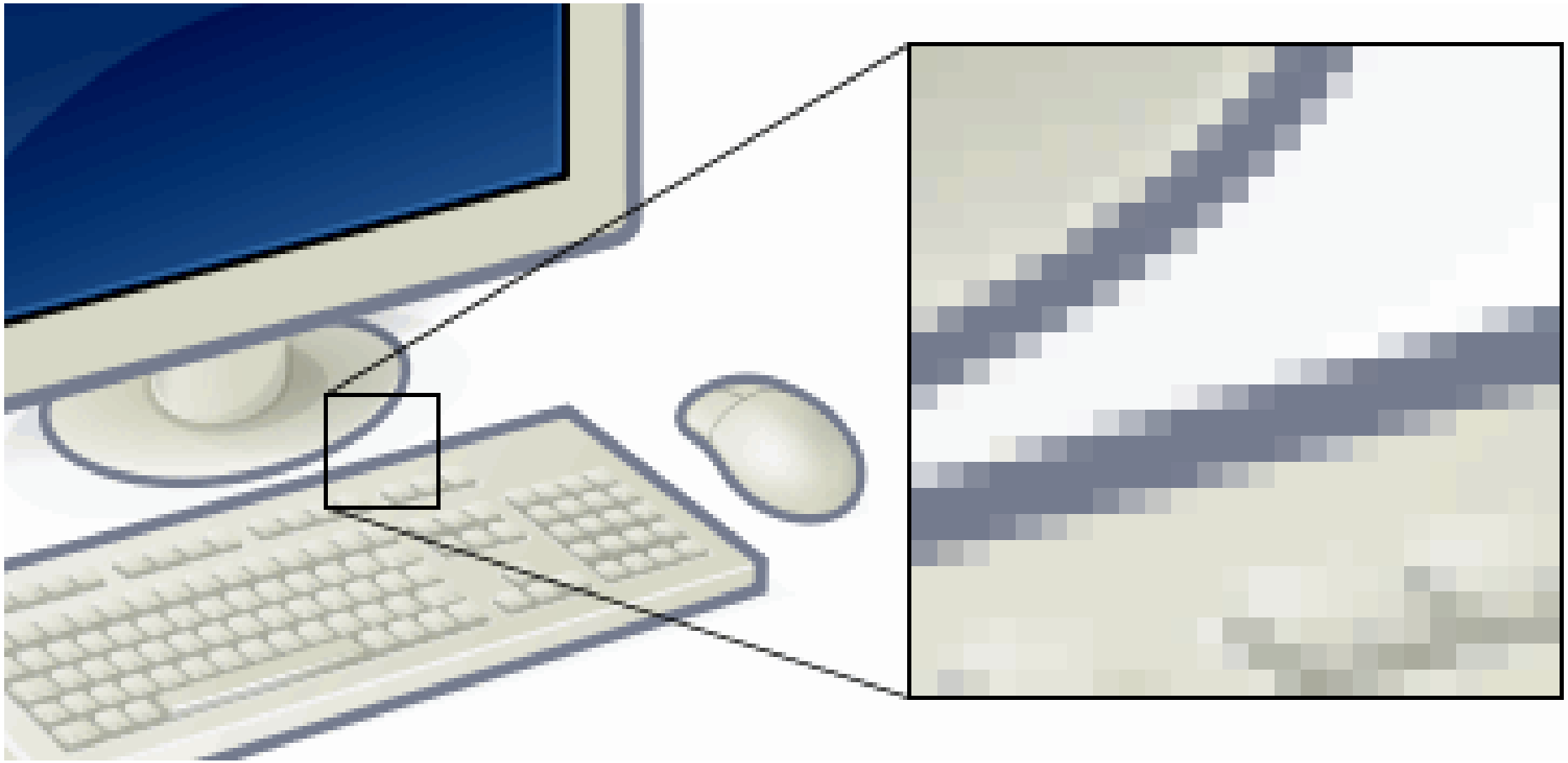


Image source: ed g2s, CC BY-SA 3.0 (https://creativecommons.org/licenses/by-sa/3.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Pixel-example.png

Quantization

From Wikipedia, page "Quantization (signal processing)":

> In mathematics and digital signal processing, **quantization** is the process of mapping input values from a large set (often a continuous set) to output values in a (countable) smaller set, often with a finite number of elements. Rounding and truncation are typical examples of quantization processes.

Quantization

Analog-digital conversion with 2-bit integers:

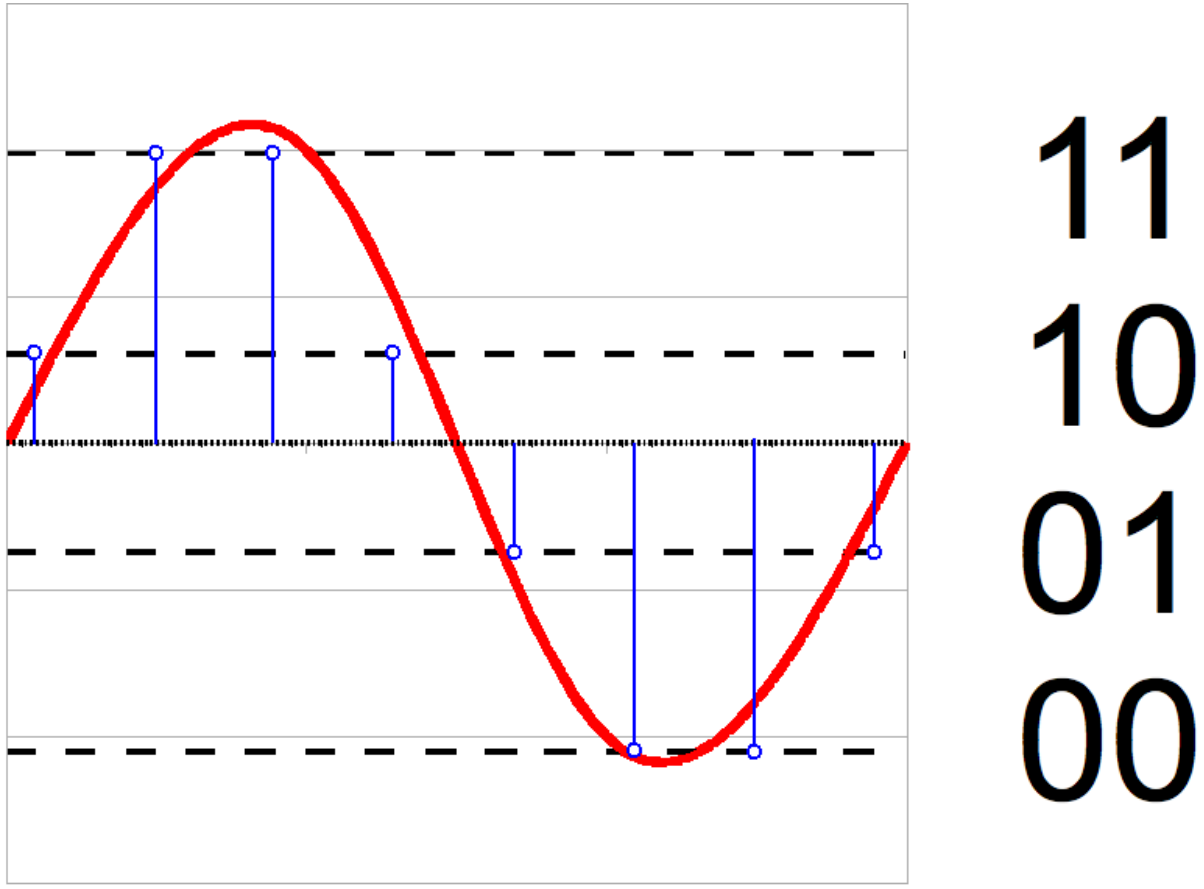


Image source: Hyacinth, CC BY-SA 3.0 (https://creativecommons.org/licenses/by-sa/3.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:2-bit_resolution_analog_comparison.png

Quantization

Analog-digital conversion with 3-bit integers:

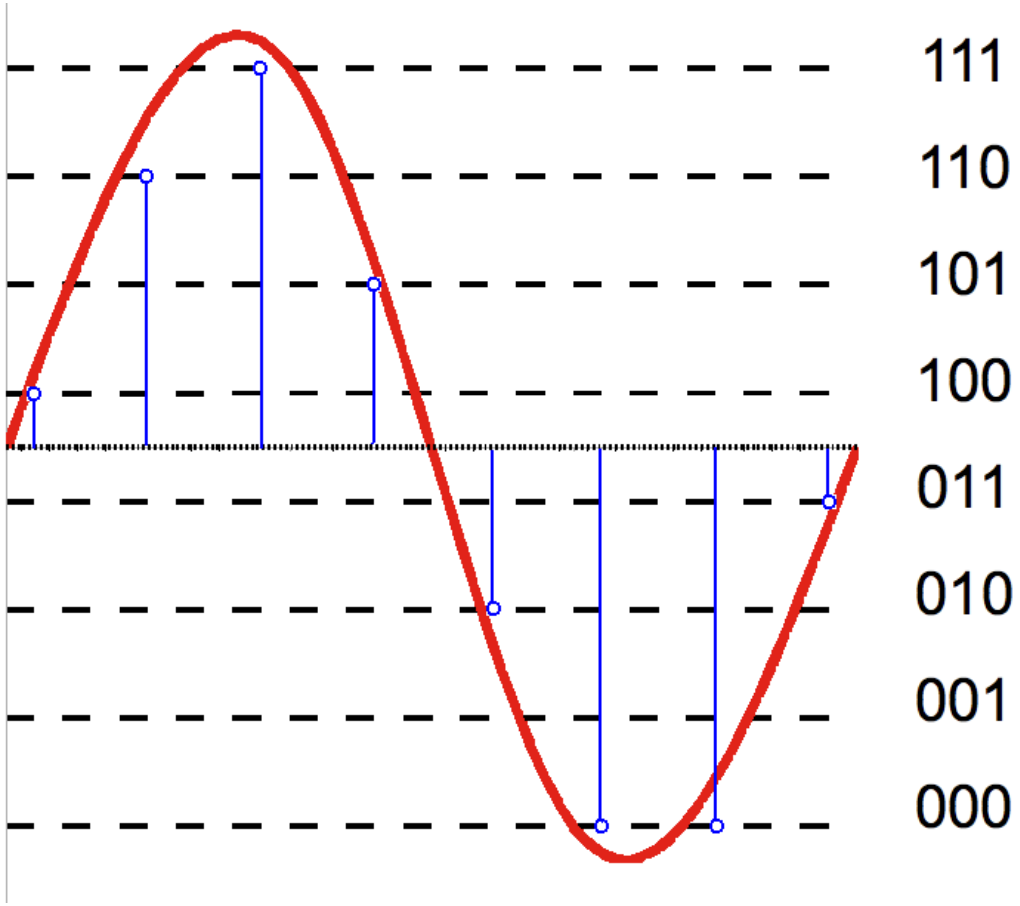


Image source: Hyacinth, CC BY-SA 3.0 (https://creativecommons.org/licenses/by-sa/3.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:3-bit_resolution_analog_comparison.png



Quantization

Quantization represents the amplitude of the continuous signal at a given sampling point to a digital value. This involves rounding or truncation and thus results in a loss of information.

Image source: https://www.pexels.com/photo/wall-blur-dangerous-colors-1679696/

Integer values

Integer values in computer science represent mathematical integer numbers by converting them to base two for digitization. This means the step size is constant at 1 and only the representable value range differs with the number of bits:

- 8-bit integers allow for $2^8 = 256$ values.
- 16-bit integers allow for $2^{16} = 65536$ values.
- 32-bit integers allow for $2^{32} = 4294967296$ values.
- 64-bit integers allow for $2^{64} = 18446744073709551616$ values.

These are unsigned ranges. Signed integers represent half positive, half negative values (e.g., 8-bit signed: –128 to +127).

Floating-point values

Floating-point values in computer science represent mathematical real numbers in an exponential format:

$N = S \times B^E$ where N is the represented number, S is the significand, B is a fixed base and E is the exponent.

Floating-point values

An example from Wikipedia, page "Floating-point arithmetic":

$$\frac{2469}{200} = 12.345 = 12345 \times 10^{-3}$$

Floating-point values

The significand S and exponent E can in a computer be represented as integer values with a fixed number of bits.

For example, in the *IEEE 754* standard, a 32-bit floating-point value uses 24 bits for the significand (including a 1 bit sign) and 8 bits for the exponent or 11 bits for the significand and 5 bits for the exponent in a 16-bit floating-point value.

Floating-point values

Floating-point values use an exponential representation, which means the step size is not fixed and floating-point values have a higher precision around the value 0. The green lines indicate representable floating-point values:



Image source: Joeleoj123, CC BY-SA 4.0 (https://creativecommons.org/licenses/by-sa/4.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:FloatingPointPrecisionAugmented.png

Integer vs. floating-point values

Use integer when:

- Signal contains only integer numbers.
- Constant rounding error, due to equal step sizes.
- Hardware efficiency is needed.

Use floating-point when:

- Signal contains real numbers.
- High value range necessary.
- High precision around zero necessary.

Quantization error

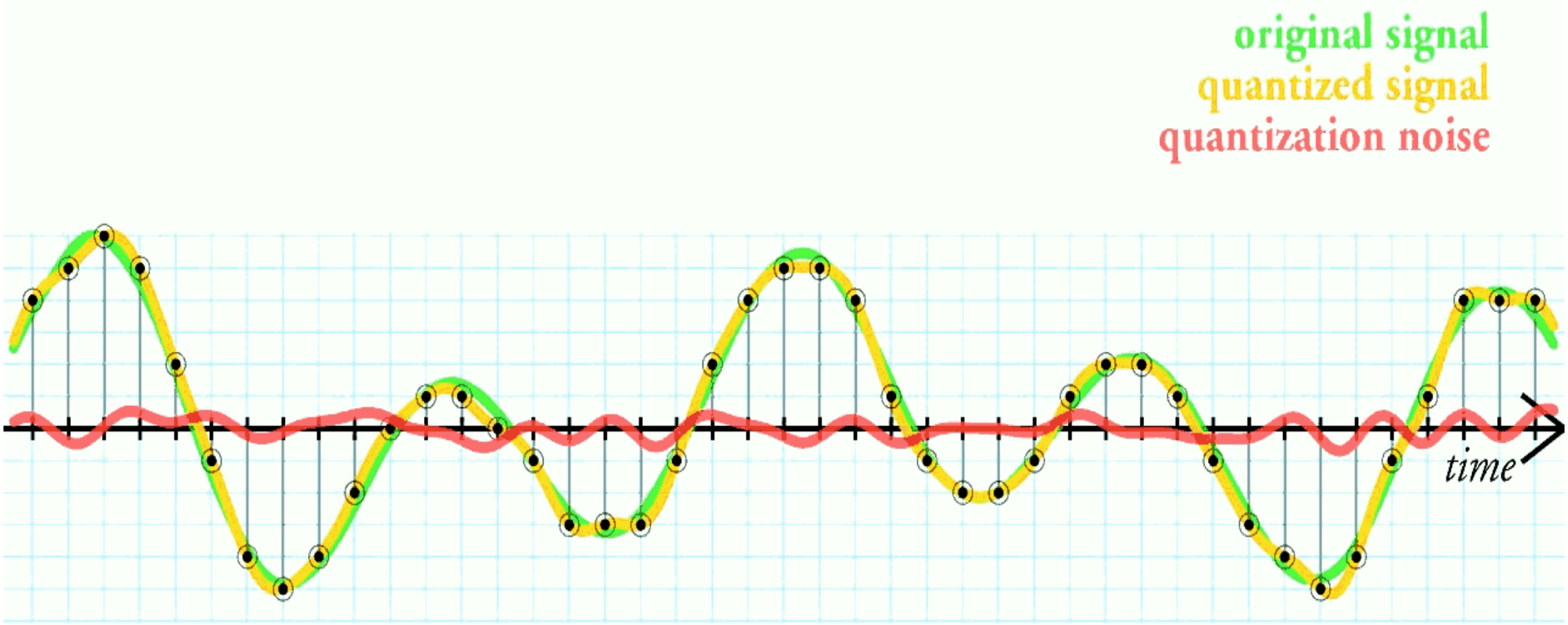


Image source: Gregory Maxwell, CC BY 3.0 (https://creativecommons.org/licenses/by-sa/3.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Quantization_error.png

Quantization error

Quantization error or quantization noise is the difference between the original signal and the quantized signal. It can be reduced by:

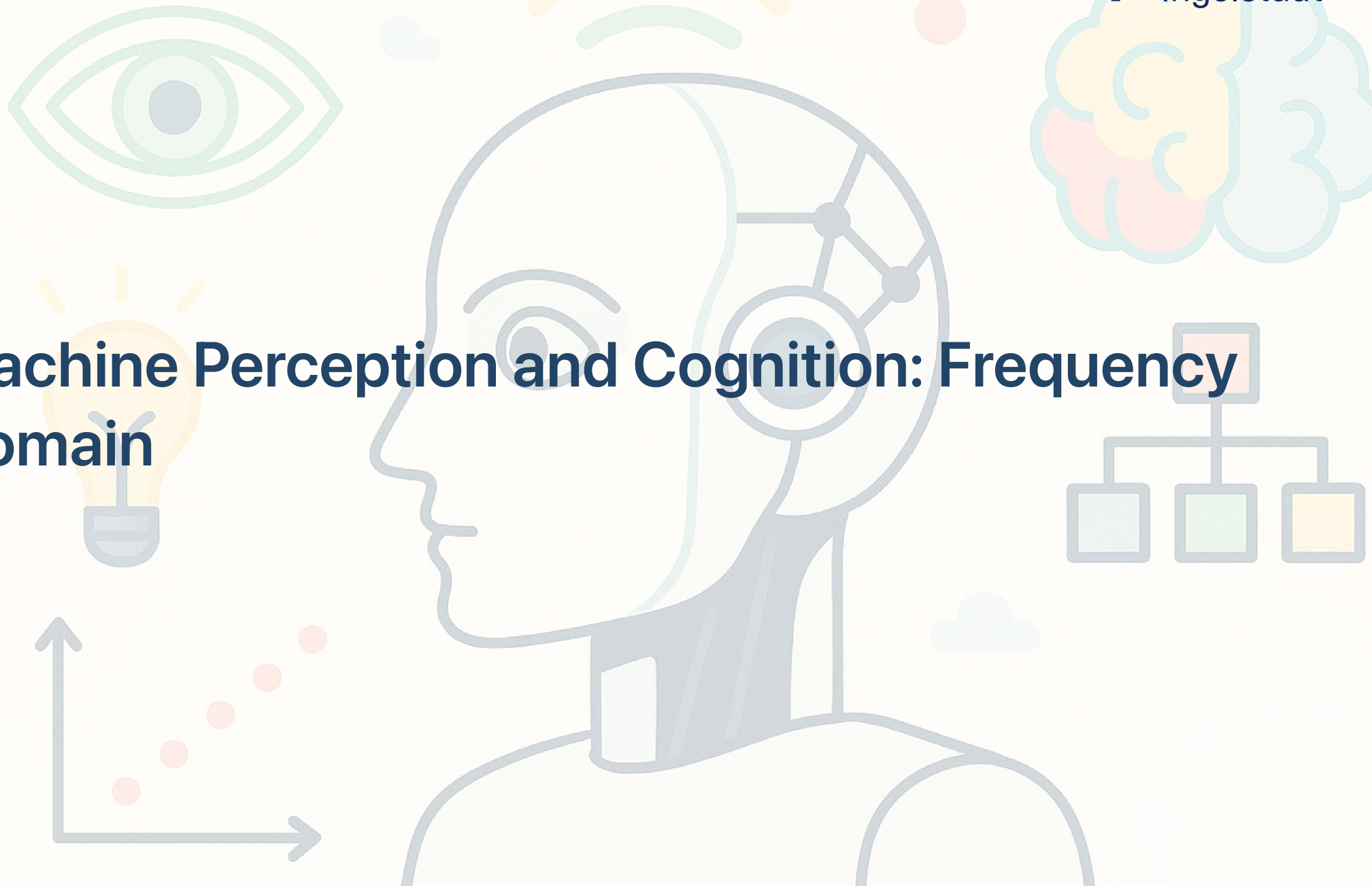
- Increase the resolution (number of bits) of the quantization.
- Center the signal on 0 when using floating-point values.
- Oversampling followed by a low-pass filter.

Sampling and quantization

Recap!

- 📊 Sampling discretizes the time/space dimension of a continuous signal.
- 📉 Quantization discretizes the amplitude of a continuous signal.
- ⏱️ Sampling is done at a constant frequency, that is equally spaced intervals.
  - This is a restriction in this course, in general there is non-uniform sampling.
- 🔢 Quantization results in integer or floating-point values.
- ⚠️ Quantization error occurs because of rounding or truncation.

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Machine Perception and Cognition: Frequency Domain

Frequency domain

So far, we have talked about continuous signals in **time domain**, that is the signal is represented as a change over time. Another representation is in **frequency domain**, that is a change over different frequencies.

Frequency domain

From Wikipedia, page "Frequency domain":

> In mathematics, physics, electronics, control systems engineering, and statistics, the **frequency domain** refers to the analysis of mathematical functions or signals with respect to frequency (and possibly phase), rather than time, as in time series. While a time-domain graph shows how a signal changes over time, a frequency-domain graph shows how the signal is distributed within different frequency bands over a range of frequencies.

Fourier transformation

Any suitable continuous-time signal (for example, one that is square-integrable) can be represented as a superposition (integral or linear combination) of sinusoidal components. This fact underlies the Fourier transform, which represents a continuous-time signal in the frequency domain instead of the time domain.

Fourier transformation

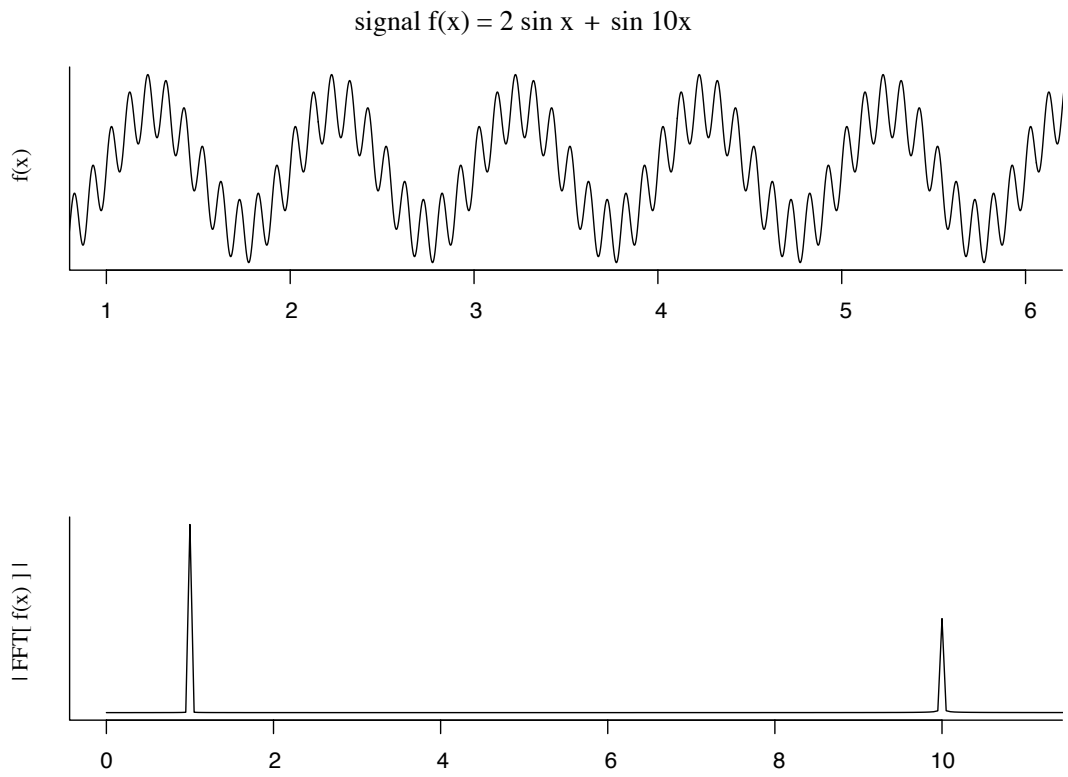


Image source: Accountalive, CC0, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Simple_time_domain_vs_frequency_domain.svg

Fourier transformation

Fourier transformation as animation with time domain in red and frequency domain in blue (animated):



Image source: Lucas Vieira, Public domain, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Fourier_transform_time_and_frequency_domains.gif

Fourier transformation

C-major chord, consisting of notes C+E+G, on a piano transformed to frequency domain:

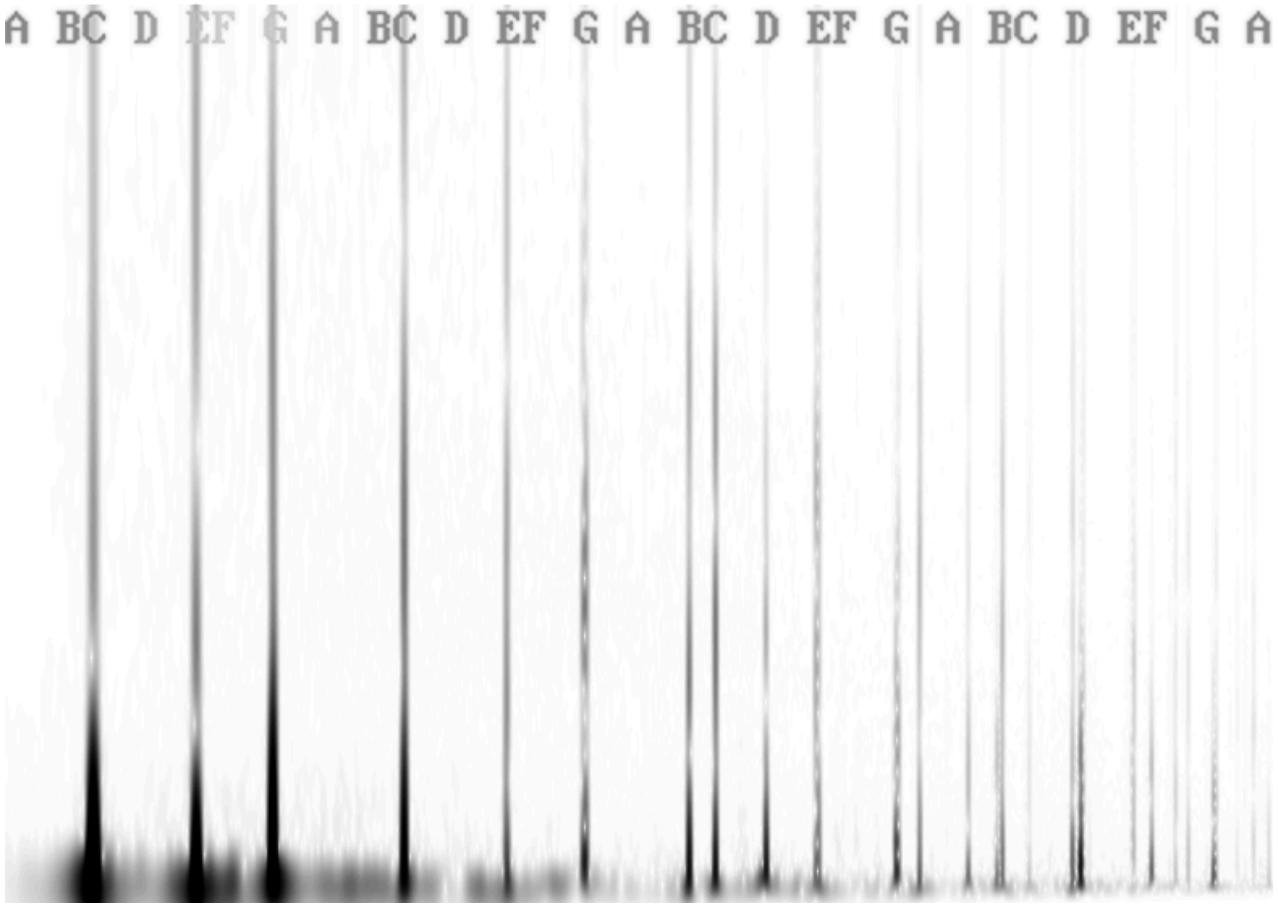


Image source: Em3rgent0rdr, CC BY-SA 4.0 (https://creativecommons.org/licenses/by-sa/4.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:CQT-piano-chord.png



Fourier transformation

But what to do with it?

Image source: https://www.pexels.com/photo/question-mark-on-crumpled-paper-5428826/

Reducing the quantization error

quantized signal = original signal + quantization noise

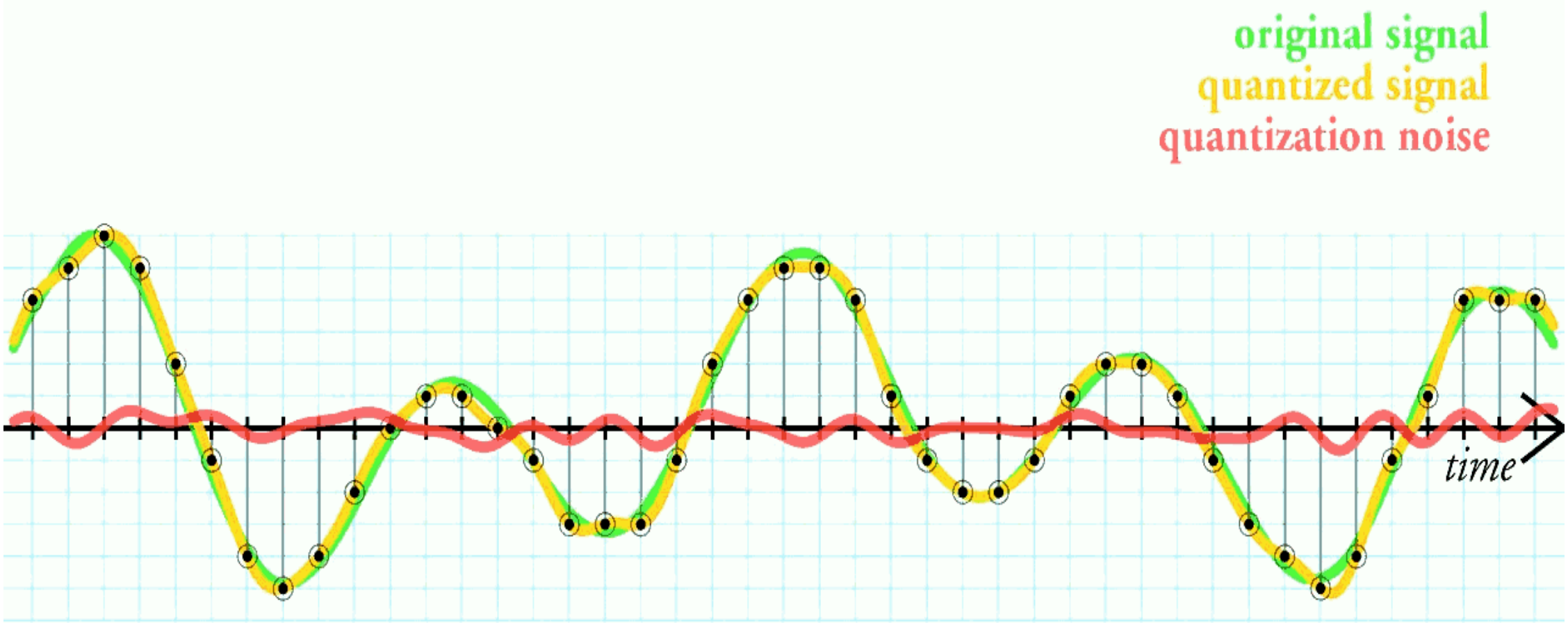


Image source: Gregory Maxwell, CC BY 3.0 (https://creativecommons.org/licenses/by-sa/3.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Quantization_error.png



Reducing the quantization error

Observe that the original signal is actually of low frequency (long wavelength), but the quantization error is of a high frequency (short wavelength). The quantized signal is the sum of the two.

Image source: https://www.pexels.com/photo/clear-light-bulb-placed-on-chalkboard-355952/

Reducing the quantization error

Quantization error can be reduced this way:

- Oversample the signal with a higher sampling frequency than frequencies occurring in the original signal.
- Quantization noise will thus also be of a higher frequency, if:
  - the quantization noise is uncorrelated with the signal.
  - the signal is "sufficiently busy".
- Thus, the original signal will be of low frequency and quantization noise of high frequency.
- Apply a low-pass filter to let low frequency signals pass, but remove high frequency signals.

Low-pass filter in Python

An exponential moving average acts like a first-order low-pass filter:

```
def low_pass_filter(prev_value, new_value, alpha):
    return alpha * prev_value + (1 - alpha) * new_value
```

Low-pass filter in Python

Applying the low-pass filter:

```
noisy_signal = [1, 2, 3, 2, 1, 2, 3, 2, 1] # just an example...
cleaned_signal = []
prev = noisy_signal[0]
alpha = 0.9
for value in noisy_signal:
    prev = low_pass_filter(prev, float(value), alpha)
    cleaned_signal.append(prev)
```

Nyquist-Shannon sampling theorem

How do we know what sampling frequency to choose?



Image source: https://www.pexels.com/photo/question-mark-on-crumpled-paper-5428826/

Nyquist-Shannon sampling theorem

Sine waves of different frequencies can actually lead to identical sampled signals:

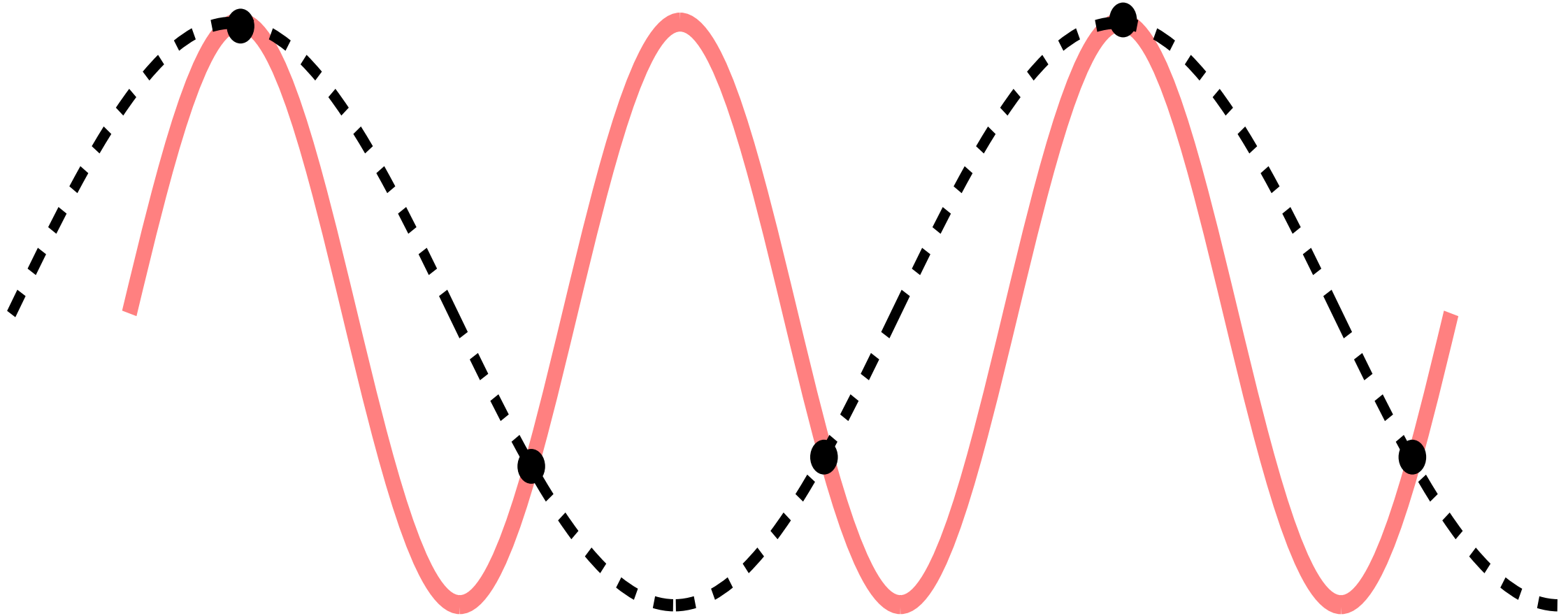


Image source: Pluke, CC0, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:CPT-sound-nyquist-thereom-1.5percycle.svg

Nyquist-Shannon sampling theorem

From Wikipedia, page "Nyquist–Shannon sampling theorem":

> The **Nyquist–Shannon sampling theorem** is a theorem in the field of signal processing which serves as a fundamental bridge between continuous-time signals and discrete-time signals. It establishes a sufficient condition for a sample rate that permits a discrete sequence of samples to capture all the information from a continuous-time signal of finite bandwidth.

Nyquist-Shannon sampling theorem

From Wikipedia, page "Nyquist–Shannon sampling theorem":

> **Theorem** — If a function *x(t)* contains no frequencies higher than *B* hertz, then it can be completely determined from its sampled values at a sequence of points spaced less than *1/(2B)* seconds apart.
> A sufficient sample-rate is therefore anything larger than *2B* samples per second.

Nyquist-Shannon sampling theorem

What sampling frequency would we need here?

Assume the frequency of the red waveform as 3 Hertz and of the black-dotted waveform as 2 Hertz.

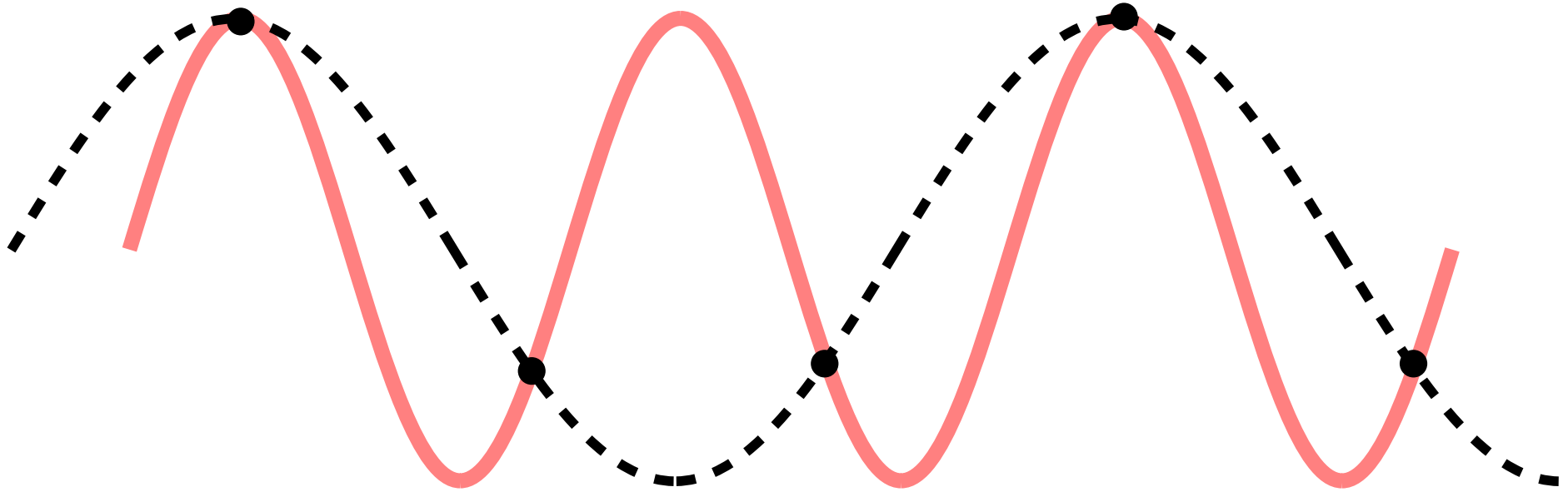


Image source: Pluke, CC0, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:CPT-sound-nyquist-thereom-1.5percycle.svg

Sampling frequency in rasterized images

The same is true for rasterized images! What image resolution would we need?

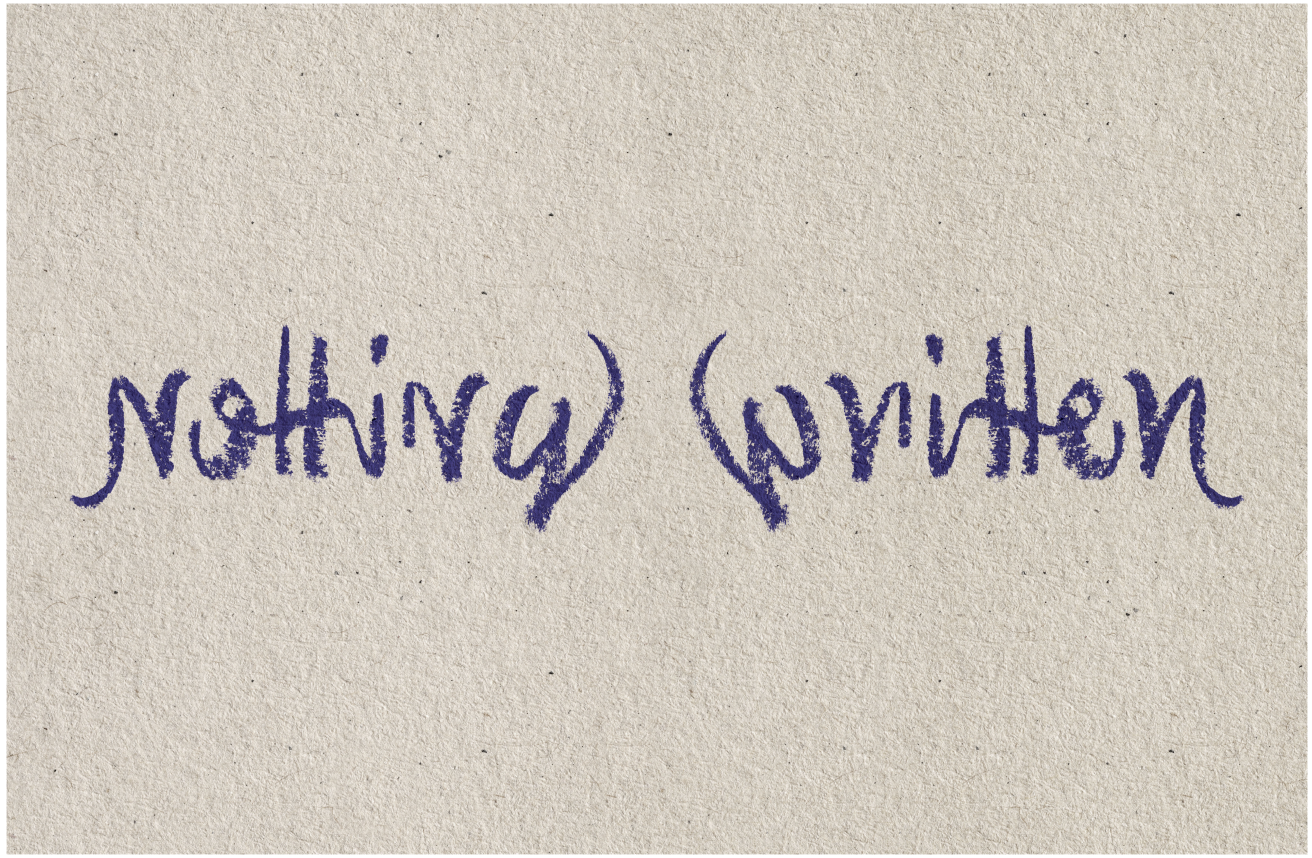


Image source: Basile Morin, CC BY-SA 4.0 (https://creativecommons.org/licenses/by-sa/4.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Ambigram_Nothing_written.jpg



Sampling frequency of human hearing

The range of audible frequencies in human hearing is roughly 20 Hz to 20000 Hz. Professional audio recording equipment often records at 48000 Hz sampling rate. Why is this?

Image source: https://www.pexels.com/photo/white-compact-disc-93388/

High-resolution rasterized image!

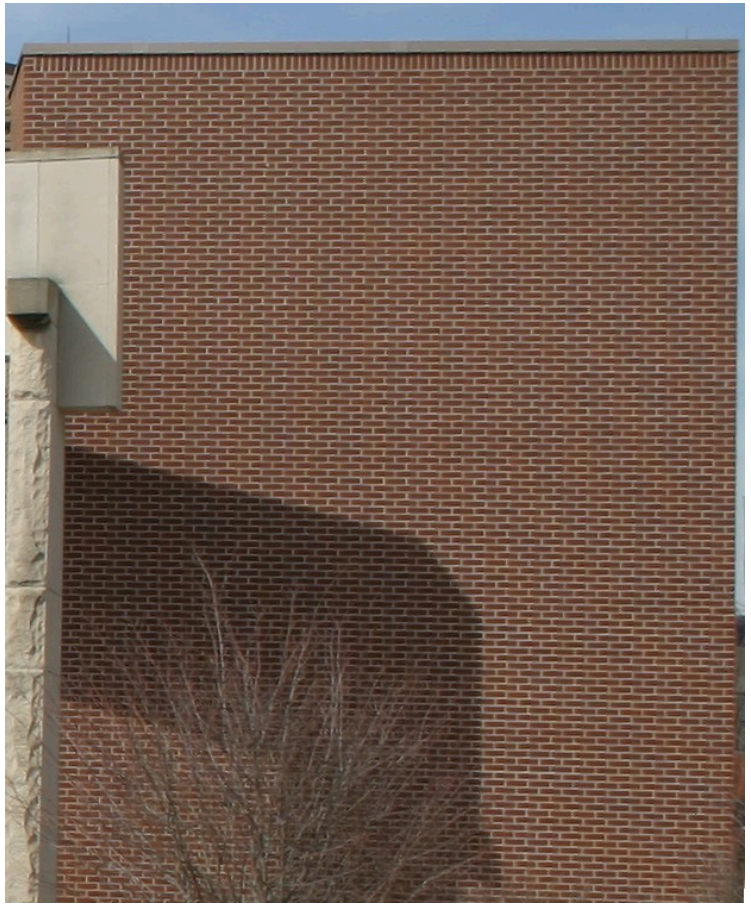


Image source: Cburnett, CC BY-SA 3.0 (https://creativecommons.org/licenses/by-sa/3.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Moire_pattern_of_bricks.jpg

Low-resolution rasterized image!

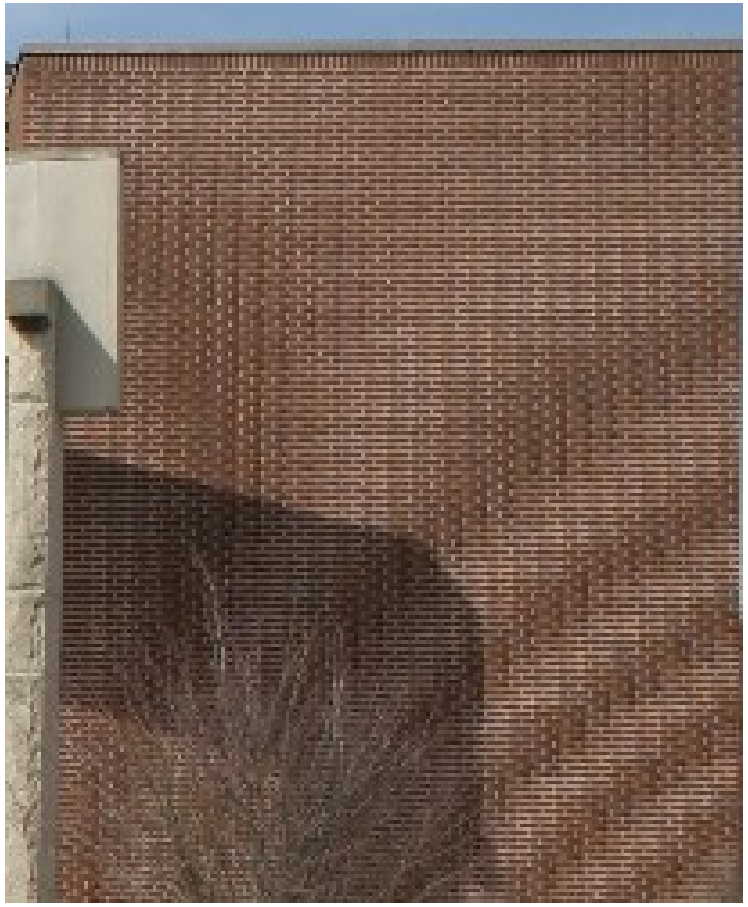


Image source: Cburnett, CC BY-SA 3.0 (https://creativecommons.org/licenses/by-sa/3.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Moire_pattern_of_bricks_small.jpg

Frequency domain

Recap!

- ⏱️ Time-domain shows the change of continuous signals over time.
- 📊 Frequency-domain shows the frequencies contained in a continuous signal.
- 🔄 Fourier transformation decomposes a time-domain signal to frequency-domain.
- 🎧 Oversampling and low-pass filtering reduces quantization noise.
- 📡 Nyquist-Shannon theorem allows us to choose a sufficient sampling frequency.